

Fiber Optic Network Design

College of Opt. Sci. & Engr., ZJU
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Fiber Raman Amplifier

- Introduction
- The Principle of Raman Effect
 - History
 - Theory
 - Structure
 - Parameters
- Simulation of FRA
 - Elements
 - Diagram
 - Result

About Raman



- Sir Chandrasekhara Venkata Raman (1888 –1970) Indian physicist
- 1930 Nobel Prize for his ground-breaking work in the field of light:
His discovery: when light traverses a transparent material, some of the deflected light changes in wavelength. --
-----Raman scattering /Raman effect.

About Raman

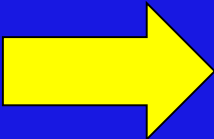


- **实地观察：**1921年夏天，航行在地中海的客轮“纳昆达”号上，有一位印度学者正在甲板上用简便的光学仪器俯身对海面进行观测。他对海水的深蓝色着了迷，一心要追究海水颜色的来源。这时他**33**岁。
- **前人结果：**熟悉著名物理学家瑞利用分子散射中散射光强与波长四次方成反比的定律（也叫瑞利定律）对蔚蓝色天空所作的解释。
- **深入思考：**他注意到瑞利的一段话值得商榷，瑞利说：“深海的蓝色并不是海水的颜色，只不过是天空蓝色被海水反射所致。”

About Raman



- 瑞利对海水蓝色的论述一直是拉曼关心的问题。他决心进行实地考察。
- 他用尼科尔棱镜观察沿布儒斯特角从海面反射的光线，即可消去来自天空的蓝光。这样看到的光应该就是海水自身的颜色。
- 结果证明，由此看到的是比天空还更深的蓝色。用光栅分析海水的颜色，发现海水光谱的峰值比天空光谱的峰值更偏蓝。



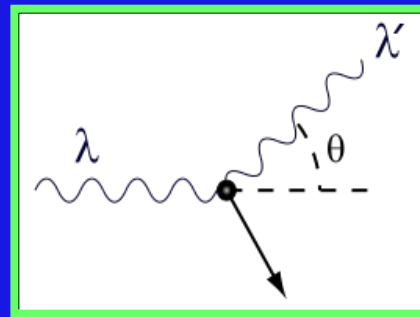
海水的颜色并非由天空颜色引起，而是海水本身的一种性质。

About Raman



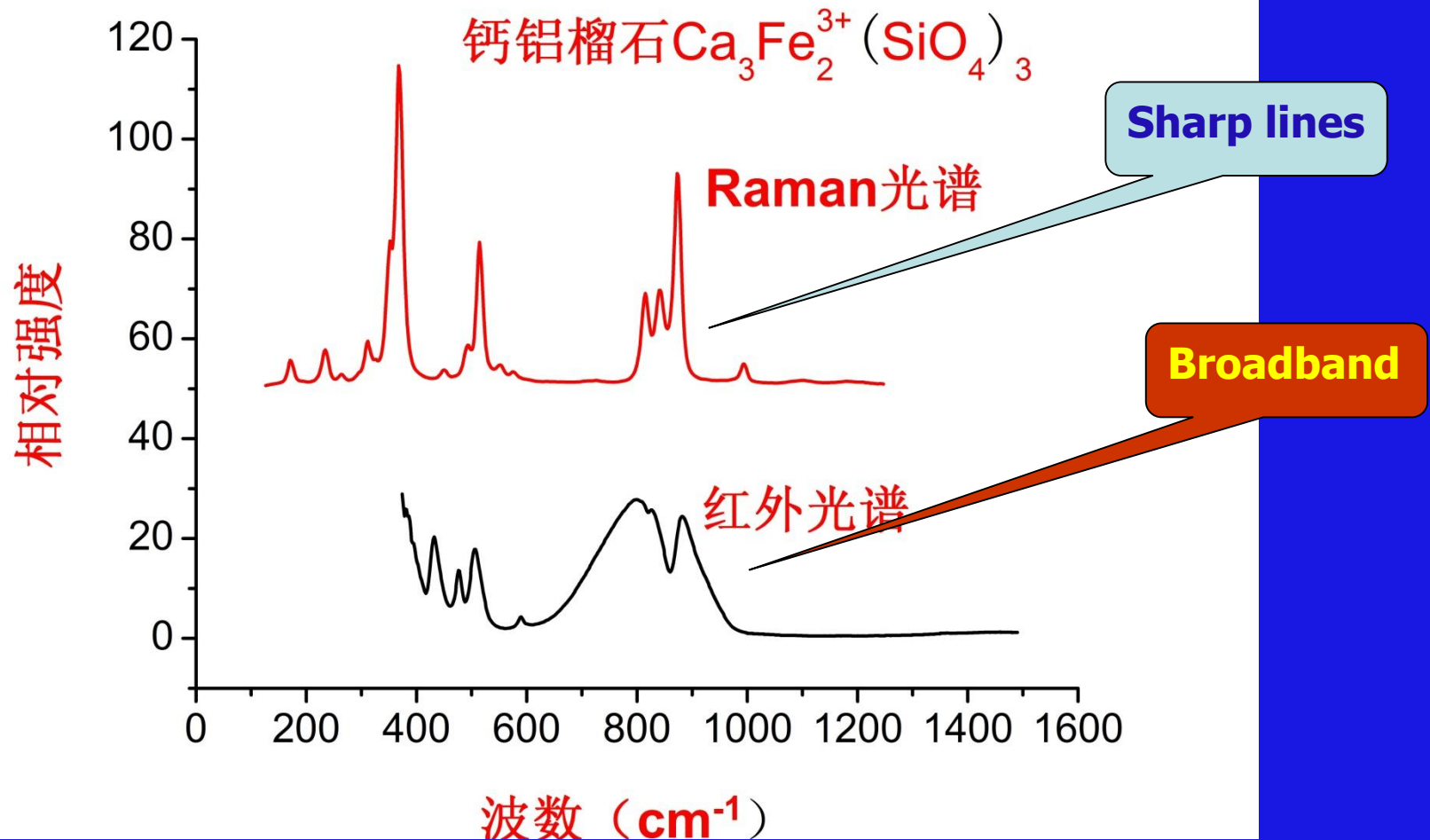
- 返回印度后，拉曼探索各种透明媒质中光散射的规律。考察各种媒质分子散射时所遵循的规律，
 - 选取不同的分子结构、
 - 不同的物态、
 - 不同的压强和温度，相变临界点等，进行散射实验。
- **1923**年他的学生拉玛纳桑（**K. R. Ramanathan**）第一次观察到了光散射中颜色改变的现象。实验是以太阳作光源，经紫色滤光片后照射盛有纯水或纯酒精的烧瓶，然后从侧面观察，却出乎意料地观察到了很弱的绿色成份。（杂质弱荧光？）
- 拉曼的学生克利希南（**K. S. Krishnan**）观测了经过提纯的**65**种液体的散射光，证明都有类似的“弱荧光”，
- **关键发现：**颜色改变了的散射光是部分偏振的。然而，荧光是一种自然光，不具偏振性。由此证明，这种波长变化的现象不是荧光效应。

About Raman



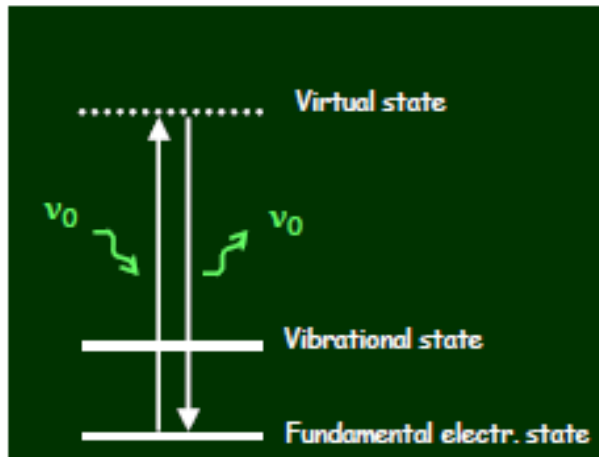
- **1923年 A.H.Compton**发现**X**射线散射后波长变长的效应（**1927**年诺奖者），而怀疑者正在挑起一场争论。拉曼从康普顿的发现得到了重要启示：莫非是“康普顿效应的光学对应”？
- 改进滤光方法：发现展现的光谱在**散射光**和入射光之间，**隔有一道暗区！！**
- **1928**年单色光判决实验。他从目测分光镜看散射光，看到在蓝光和绿光的区域里，有两根以上的尖锐亮线。每一条入射谱线都有相应的散射线。一般情况，散射线的频率比入射线低，偶而也观察到比入射线频率高的散射线，但强度更弱些。
- **1930**年，美国光谱学家 **R.W.Wood**对频率变低的散射线取名为斯托克斯线；频率变高的为反斯托克斯线。

Comparison between Raman and infrared spectroscopy

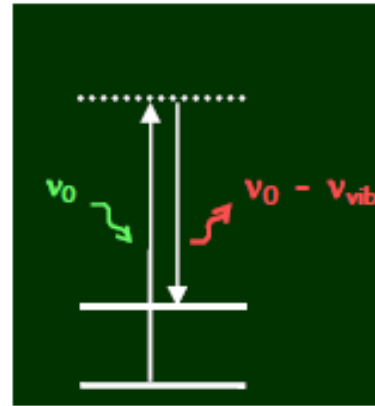


The Theory of Raman Effect

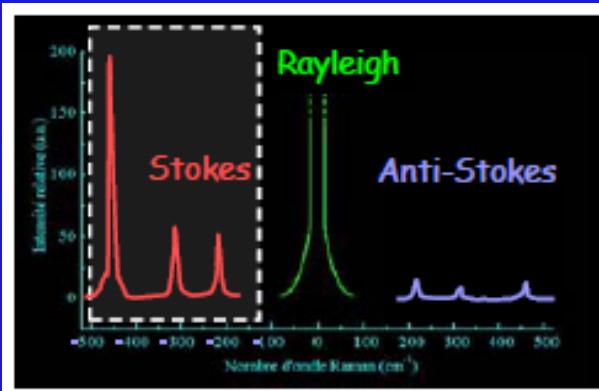
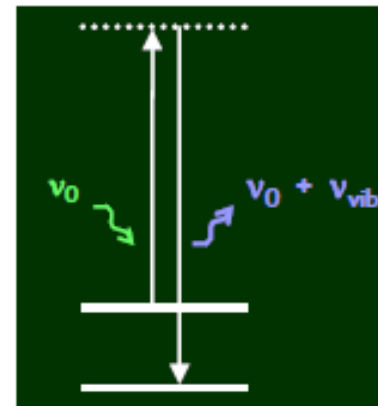
Raleigh scattering (elastic)



Stokes Scattering



Anti-Stokes scattering



$$P = \epsilon_0 \chi E$$

Light-Matter Interaction

$$\chi = \chi_0 + \left(\frac{\partial \chi}{\partial Q}\right)_0 Q + \frac{1}{2!} \left(\frac{\partial^2 \chi}{\partial Q^2}\right)_0 Q^2 + \frac{1}{3!} \left(\frac{\partial^3 \chi}{\partial Q^3}\right)_0 Q^3 + \dots$$

$$\chi(t) = \chi_0 + \left(\frac{\partial \chi}{\partial Q}\right)_0 Q_0 \cos \omega_q t$$

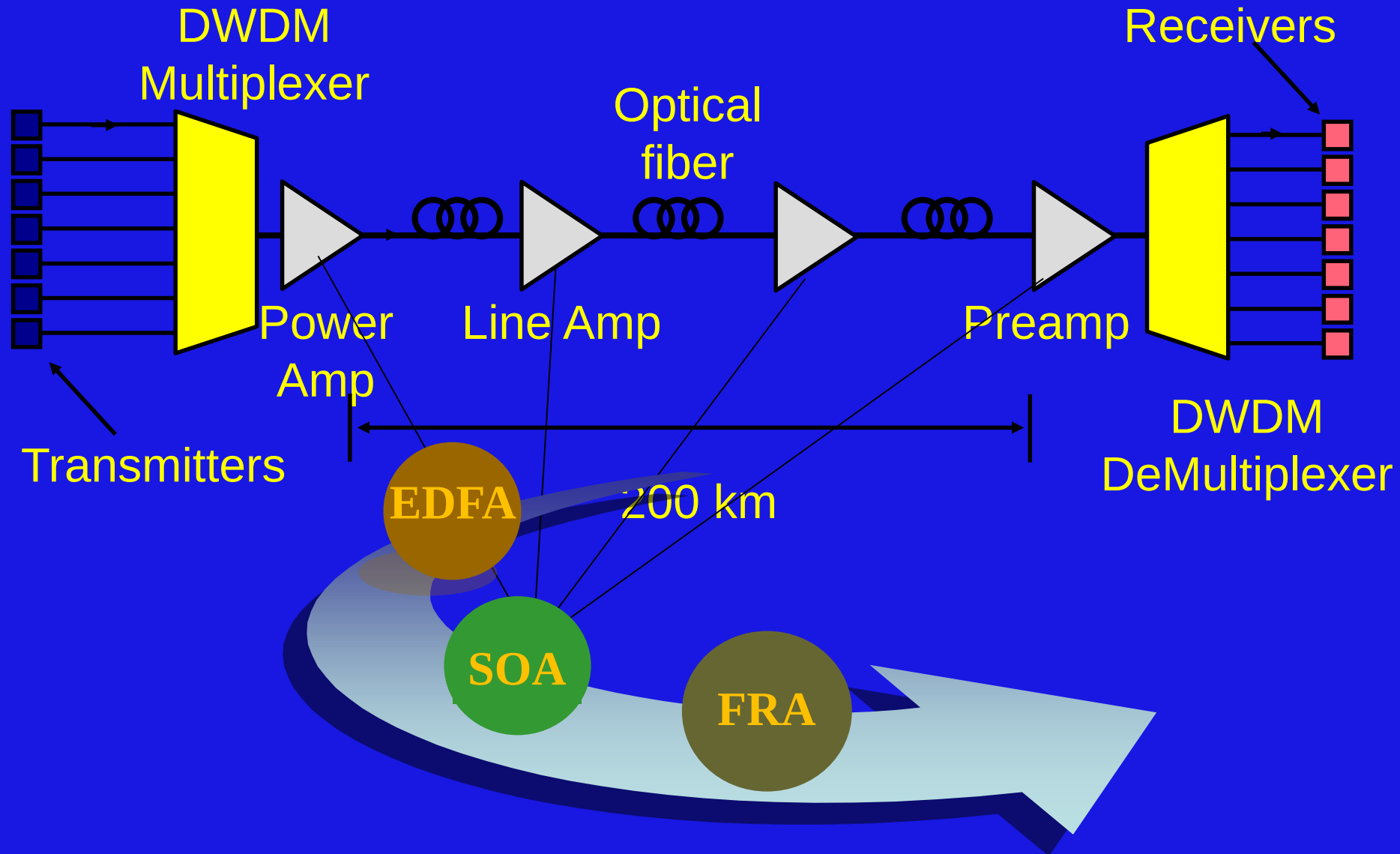
$$\begin{aligned} P(t) &= \epsilon_0 \chi_0 E_0 \cos \omega_p t + \left(\frac{\partial \chi}{\partial Q}\right)_0 Q_0 \epsilon_0 E_0 \cos \omega_p t \cos \omega_q t \\ &= \epsilon_0 \chi_0 E_0 \cos \omega_p t + \frac{1}{2} \left(\frac{\partial \chi}{\partial Q}\right)_0 Q_0 \epsilon_0 E_0 [\cos(\omega_p - \omega_q)t + \cos(\omega_p + \omega_q)t] \end{aligned}$$

The Theory of Raman Effect

Big question: Virtual State

- A virtual state is an intermediate “imaginary” state, which mediates otherwise forbidden transitions.
- not eigenfunctions of anything, parameters such as occupation, energy and lifetime need to be qualified.
- No measurement of a system will show one to be occupied, still have.
- each virtual state has an associated energy, no direct measurement of its energy is possible.
- A general concept, used to predict and describe results in many areas: Raman, non-linear optics, photochemistry, and nuclear processes.

Optical amplifiers in the OFC system



Performance

Performance comparison of optical fiber amplifiers

Type	Wavelength (nm)	Bandwidth (THz)	Saturation power (dBm)	Gain (dB)	NF (dB)
SOA	800-1600	~10	~10	~30	~6
EDFA	C and L band	2~10	10~24	~40	3~4
FRA	up to pumping source	~10	~20	20~40	~3
TDFA	S band	~5	>10	~20	~6
PDFA	1280-1340	~2	>10	~30	~6

Stimulated Raman Scattering Theory

⇒ The Raman gain can be characterized by the gain coefficient g_R such that

$$g(z) = g_R I_p(z)$$

$$g(\omega) = g_R(\omega) \frac{P_p}{a_p}$$

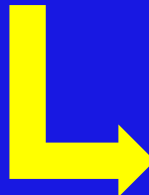
⇒ where $I_p(z)$ is the pump intensity at position z in the fiber, P_p is the pump power, and a_p is the cross-sectional area of the pump beam inside the fiber. Since a_p varies with each fiber design, we use the ratio g_R/a_p as the measure of Raman gain efficiency.

⇒ Raman amplifiers can be forward or reverse pumped. For forward pumped, the coupled power equations can be written for the pump and signal power (P_s , P_p)

$$\frac{dP_s}{dz} = -\alpha_s P_s + \left(\frac{g_R}{a_p} \right) P_p P_s$$

$$\frac{dP_p}{dz} = -\alpha_p P_p - \left(\frac{\omega_p}{\omega_s} \right) \left(\frac{g_R}{a_p} \right) P_s P_p$$

**Signal Power increases,
Pumping power decreases
with z**



$$P_p(z) = P_p(0) \exp(-\alpha_p z)$$

$$P_s(L) = P_s(0) \exp\left(\frac{g_R P_0 L_{eff}}{a_p} - \alpha_s L\right)$$

$$L_{eff} = [1 - \exp(-\alpha_p L)] / \alpha_p$$

The Raman Scattering Equation

Rayleigh scattering

$$\frac{dP^{\pm}(z, \nu)}{dz} = \mp \alpha(\nu) P^{\pm}(z, \nu) \pm \gamma(\nu) P^{\mp}(z, \nu)$$

$$\pm \sum_{\xi > \nu} \left\{ \frac{g_r(\xi - \nu)}{K_{\text{eff}} A_{\text{eff}}} [P^{\pm}(z, \xi) + P^{\mp}(z, \xi)] P^{\pm}(z, \nu) + \right.$$

Spontaneous emission

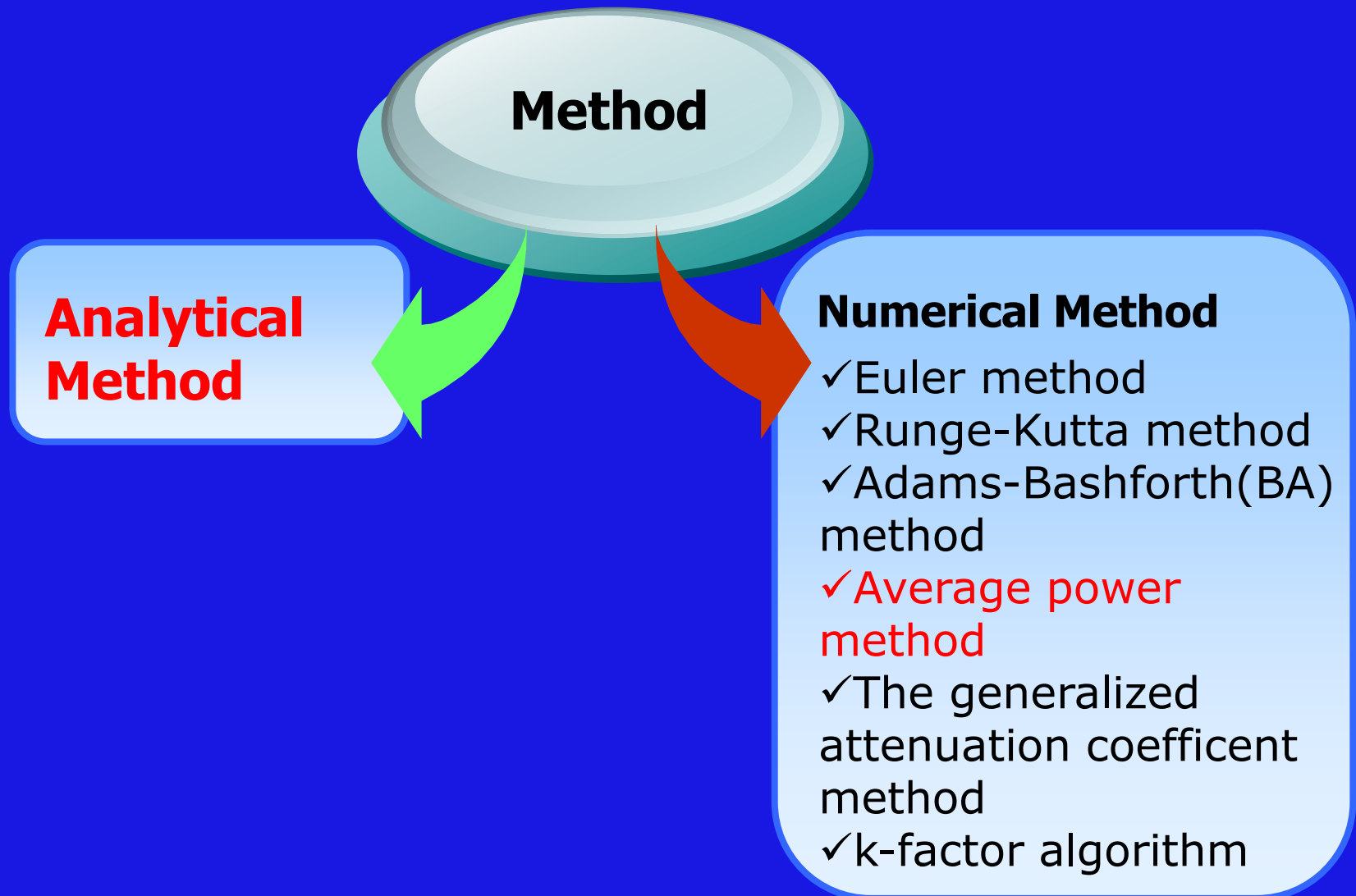
$$h\nu \frac{g_r(\xi - \nu)}{A_{\text{eff}}} [P^{\pm}(z, \xi) + P^{\mp}(z, \xi)] \left[1 + \frac{1}{e^{h(\xi - \nu)/kT} - 1} \right] \Delta \nu \Bigg\}$$

$$\mp \sum_{\xi < \nu} \frac{g_r(\nu - \xi)}{K_{\text{eff}} A_{\text{eff}}} \left(\frac{\nu}{\xi} \right) [P^{\pm}(z, \xi) + P^{\mp}(z, \xi)] P^{\pm}(z, \nu) +$$

$$2h\nu \frac{g_r(\xi - \nu)}{A_{\text{eff}}} P^{\pm}(z, \nu) \left[1 + \frac{1}{e^{h(\xi - \nu)/kT} - 1} \right] \Delta \xi \Bigg\}$$

Raman gain coefficient

Solving the Raman Scattering Equation

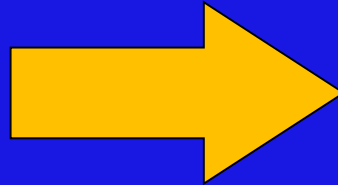


Analytical Method

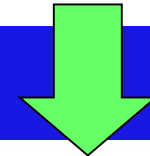
$$\frac{dP_s}{dz} = g_R P_p P_s - \alpha_s P_s$$

$$\frac{dP_p}{dz} = -\frac{\omega_p}{\omega_s} g_R P_p P_s - \alpha_p P_p$$

$$P_p = P_{p0} \exp(-\alpha_p z)$$



$$\frac{dP_s}{dz} = g_R P_{p0} \exp(-\alpha_p z) P_s - \alpha_s P_s$$



$$P_s(L) = P_s(0) \exp(g_R P_{p0} L_{eff} - \alpha_s L)$$

$$L_{eff} = [1 - \exp(-\alpha_p L)] / \alpha_p$$

For single pump and single signal

Effective length

**For multiple pump
and multiple signal**

$$P(z, \lambda) = \frac{\beta P_0^2 L_{eff} \exp\{-\alpha z\} \exp\{\beta P_0 L_{eff} (\lambda - \lambda_{min})\}}{\exp\{\beta P_0 L_{eff} (\lambda_{max} - \lambda_{min})\} - 1}$$

$$\beta = \frac{\gamma_{ik}}{2A_{eff} (\lambda_k - \lambda_i)}$$

Average Power Method

To reduce computation time, Raman amplifier is split to a cascade of small pieces. Each short piece is solved by small-signal traveling method. The average power is introduced in each short piece, so the original equation is simplified.

Backward Rayleigh scattering coefficient

Attenuation coefficient

Raman gain coefficient

Pump frequency

Stokes frequency

Backward power

Forward power

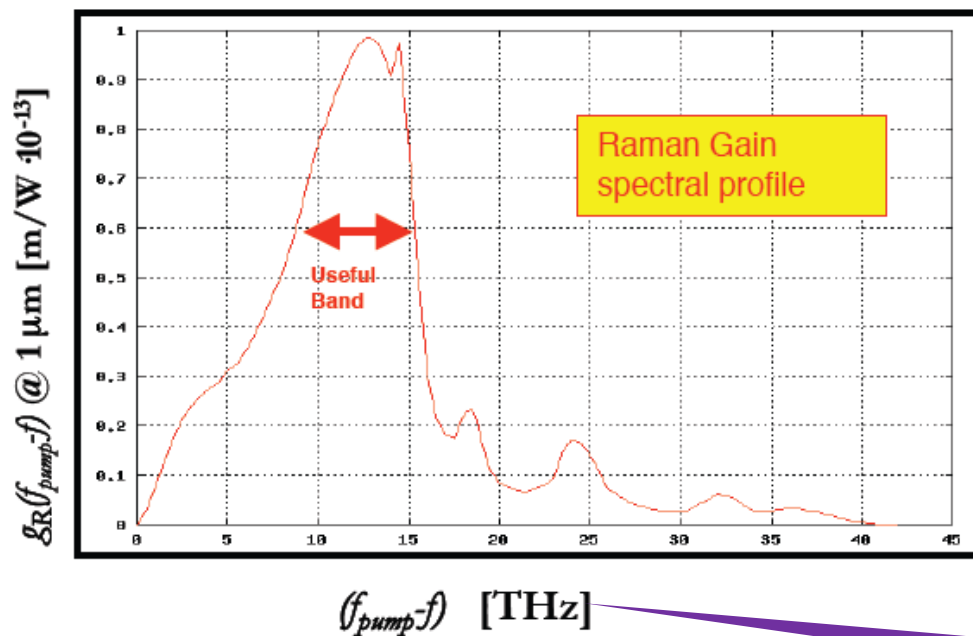
$$\begin{aligned} \frac{dP_f(z, \nu)}{dz} = & \alpha(\nu)P_f(z, \nu) + \gamma(\nu)P_b(z, \nu) + P_f(z, \nu) \sum_{\nu < \zeta} \frac{g_r(\nu - \zeta)}{K_{eff}A_{eff}} [P_f(z, \zeta) + P_b(z, \zeta)] + \\ & h\Delta\nu \sum_{\nu < \zeta} \frac{g_r(\nu - \zeta)}{A_{eff}} [P_f + P_b] [1 + \exp([h(\zeta - \nu)/kT] - 1)^{-1}] - \\ & P_f(z, \nu) \sum_{\nu > \zeta} \frac{g_r(\nu - \zeta)}{K_{eff}A_{eff}} \frac{\nu}{\zeta} [P_f(z, \zeta) + P_b(z, \zeta)] - \\ & 2h\nu\Delta\nu P_f(z, \nu) \sum_{\nu > \zeta} \frac{g_r(\nu - \zeta)}{A_{eff}} [1 + (\exp([h(\nu - \zeta)/kT] - 1)^{-1})] \end{aligned}$$

**Fiber effective
cross-section area
polarization factor**

Spontaneous emission

Gain Profile: depends on medium

- ⇒ The gain coefficient $g_r(\omega)$ has a dependence on the detuning between the pump and observation (or probe) frequency as shown below. The gain bandwidth is indicated below, and can change depending on the fiber type (e.g. dispersion compensating fiber, dispersion shifted fiber, normal single mode fiber).
- ⇒ A typical gain bandwidth defined at the FWHM of the main peak is 6THz!

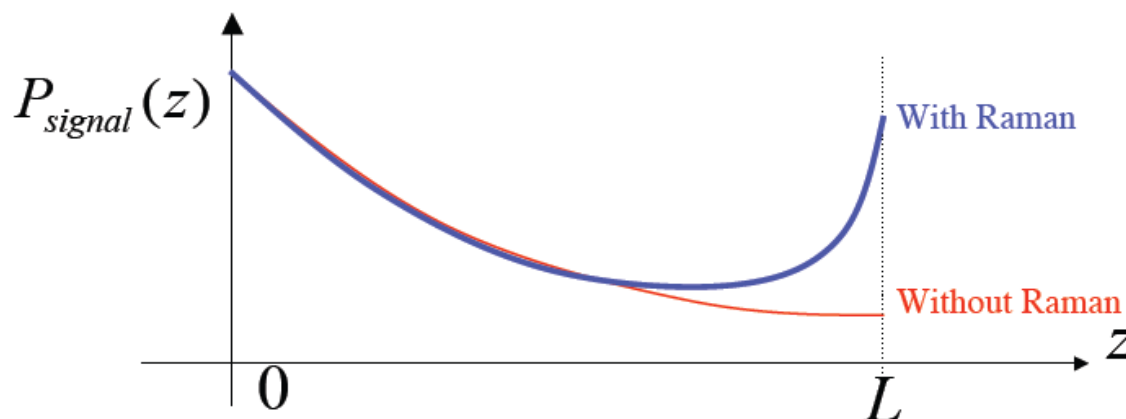
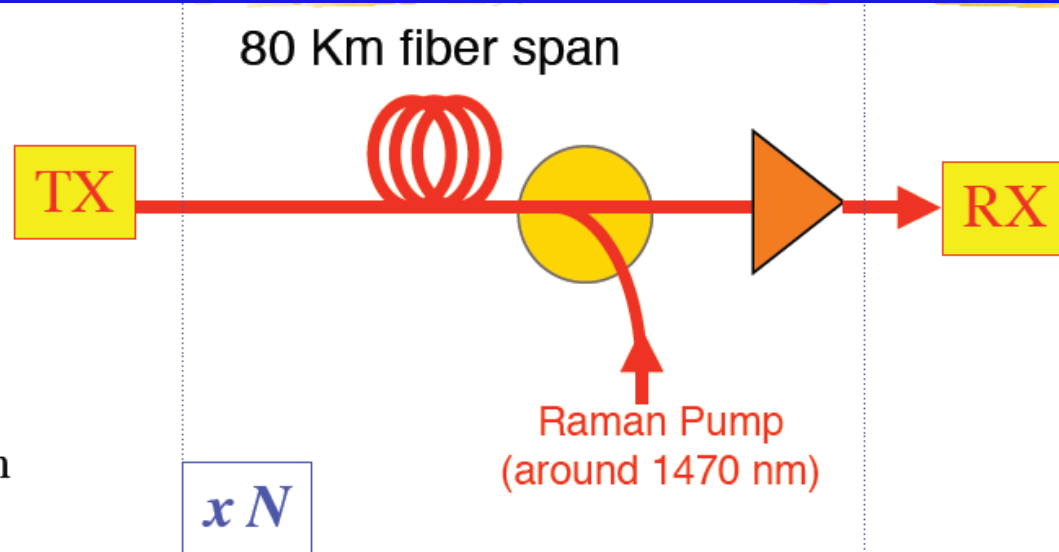


Between pumping and
signal frequency

Typical Application

⇒ The typical configuration for ultra long system is based on

- ⇒ Hybrid Raman-EDFA configuration
- ⇒ Counter-propagating Raman pumping



Bands from S to L (ITU)

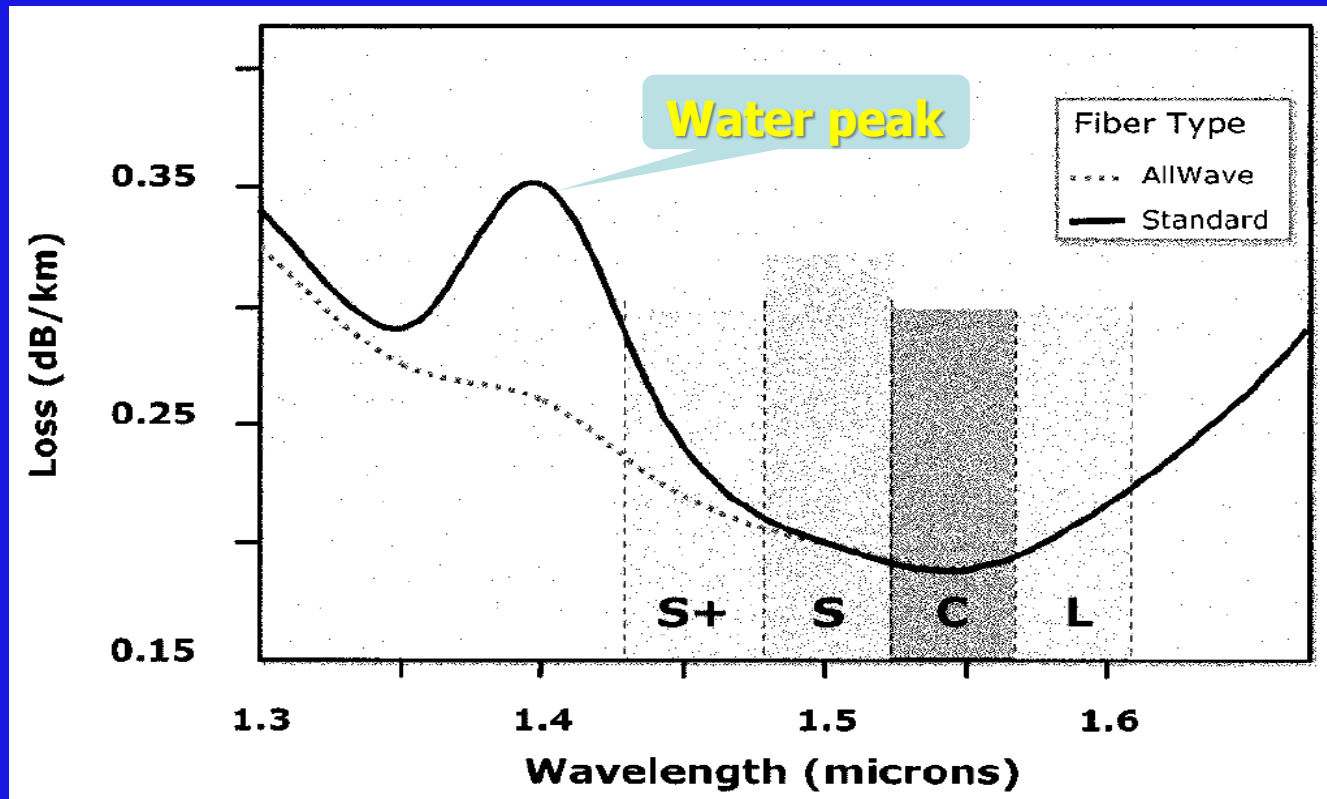
S⁺ Band: 1450~1480nm (Extended Short Band)

S⁻ Band: 1480~1530nm (Short Band)

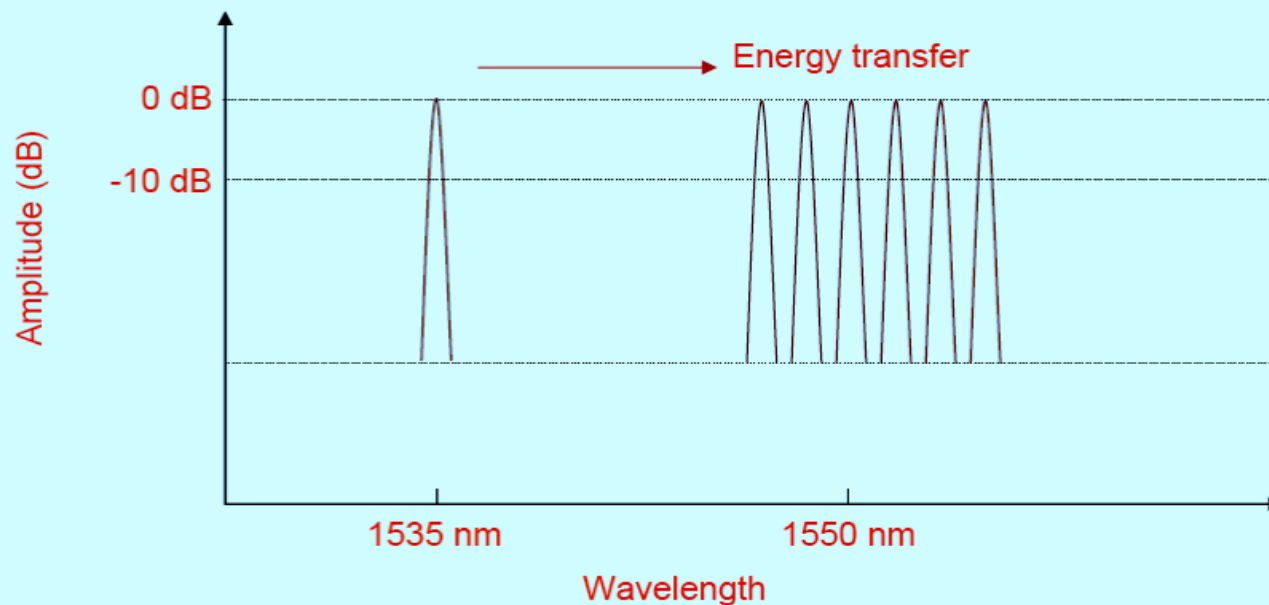
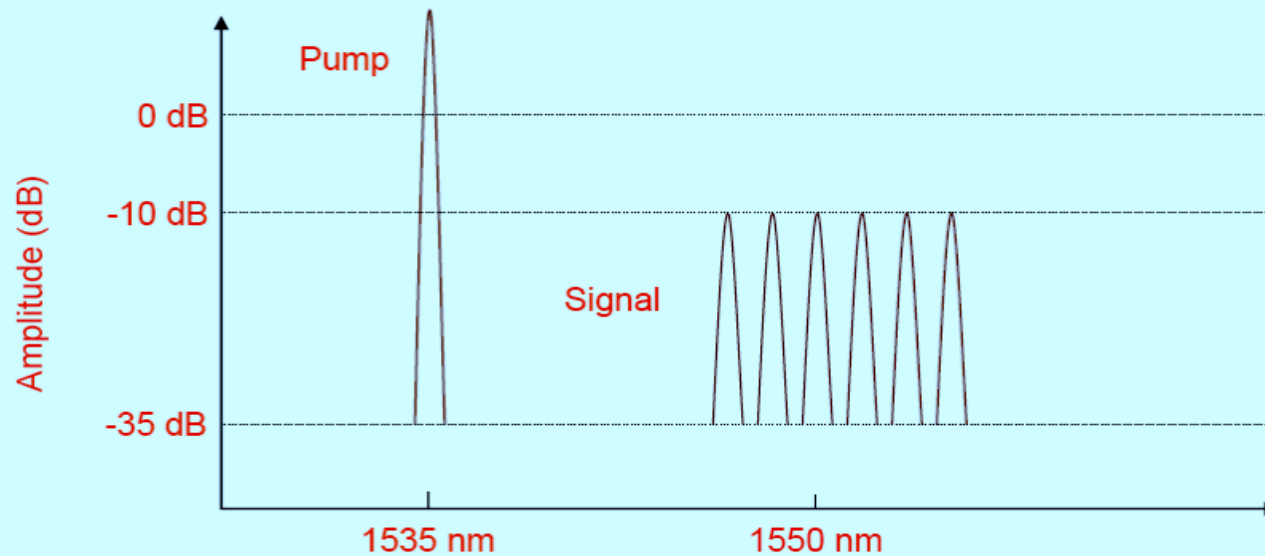
C Band : 1530~1570nm (Center Band)

L Band : 1570~1610nm (Long Band)

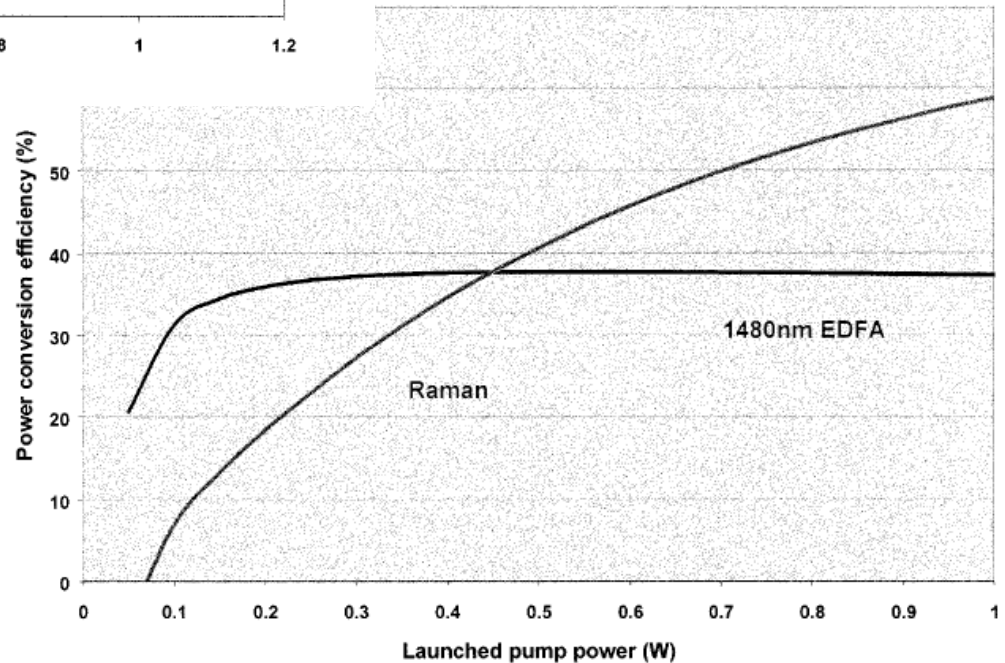
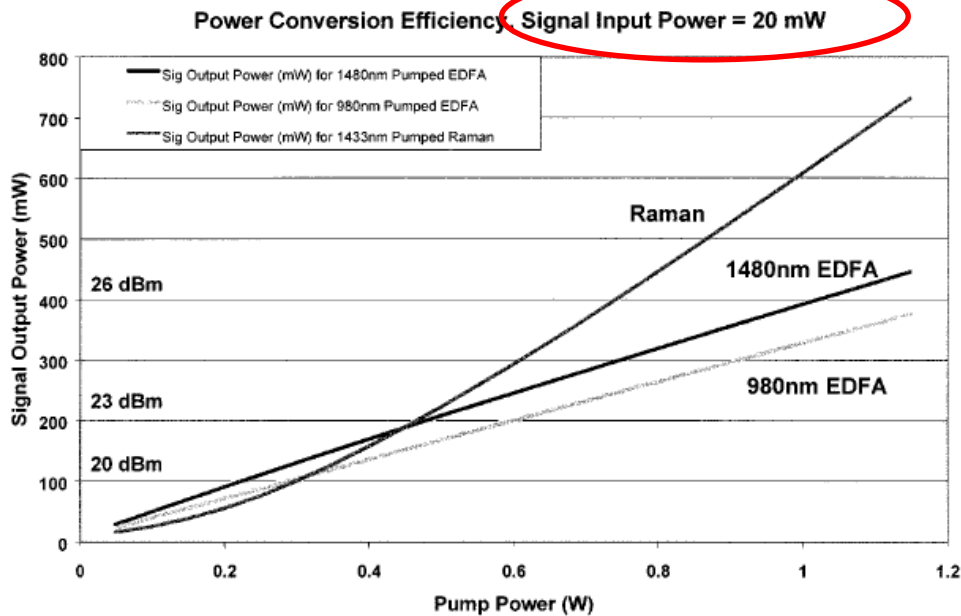
L⁺ Band : 1610~1650nm (Longer Band)



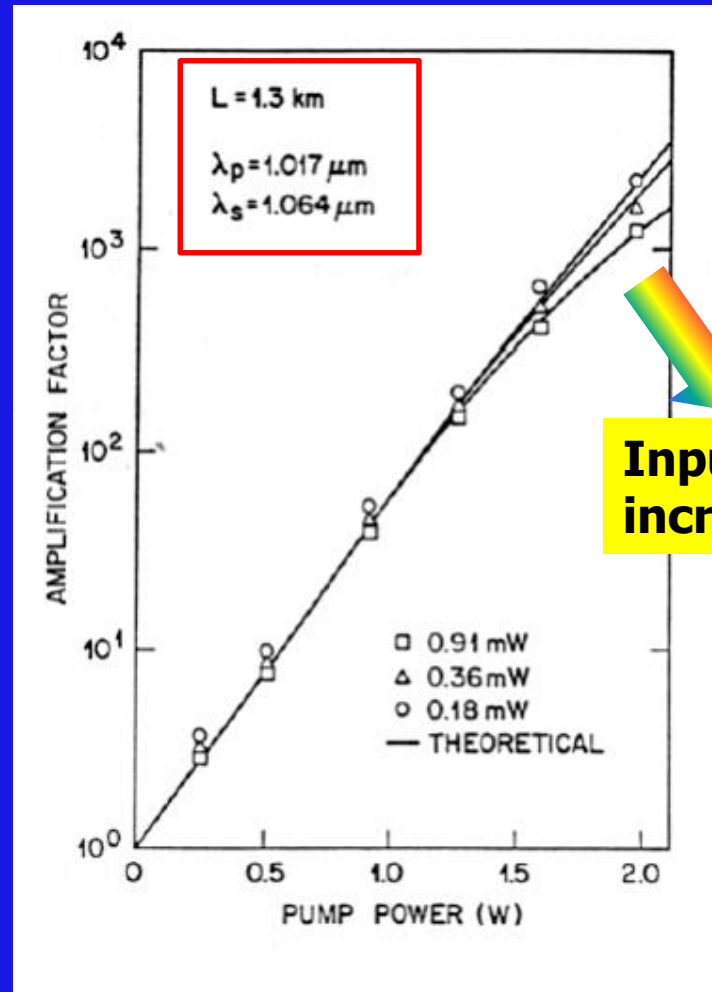
Energy Transfer in Raman Amplification



Pumping Power Conversion Efficiency



Gain Saturation



Major parameters

Pump On-off Gain

$$G_{on-off} (dB) = \log_{10} \frac{P_{on}}{P_{off}}$$

Net Gain

$$G_{net} (dB) = 10 \log_{10} \left[\frac{P_{out} - Noise_{out}}{P_{in} - Noise_{in}} \right]$$

Gain

Wavelength scanning method
Broadband spectroscopy
Laser array method

Gain Bandwidth

Half-height and full broadband width

Gain Fluctuation

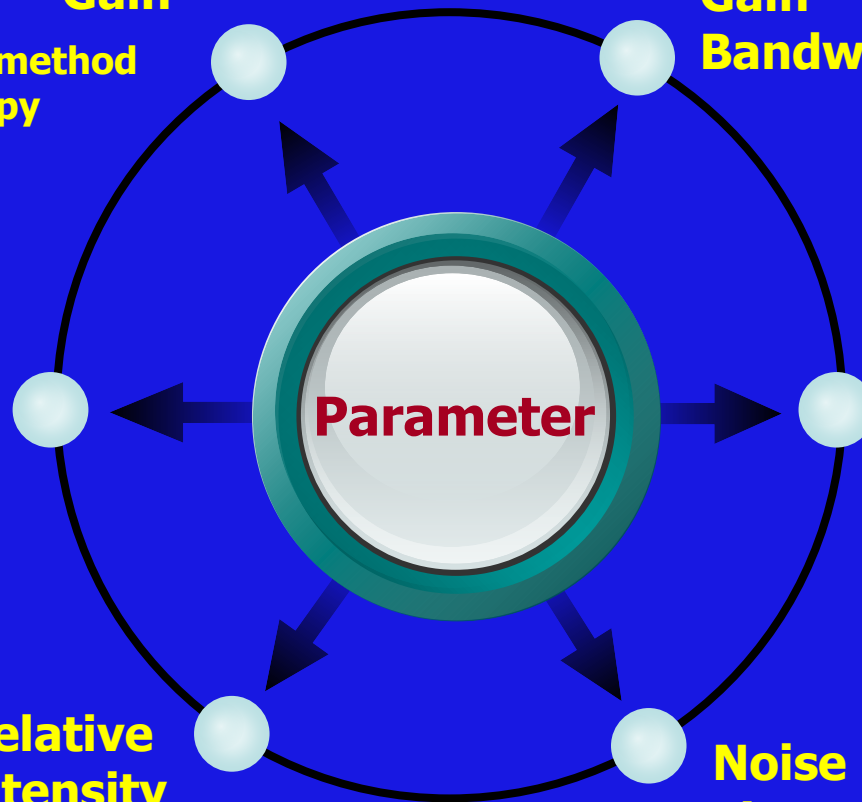
The gain difference between maximum and minimum gain wavelengths
GF < 0.5dB

Polarization Dependent gain

The random jitter of the polarization states of pump and signal light is produced by birefringence effect, which causes Raman gain jitter.

Relative intensity noise

Noise Figure



Sources of Noise in Raman Amplifiers

- **Double Rayleigh scattering (DRS)**

- Two scattering events, backward and forward, due to microscopic glass composition non-uniformity, e.g. backward ASE reflected by DRS->forward->noise

- **Short upper-state lifetime (3-6fs)**

**Special
Property!**

- Virtually instantaneous gain: pump fluctuation couples to signal->RIN

- **ASE**

- **Phonon-stimulated optical noise**

- Thermally induced phonons in fiber that can spontaneously experience gain from the pumps, thereby creating noise for signals close to pump wavelengths

Relative Intensity Noise

Laser relative intensity noise is defined as the ratio of the noise and the output light intensity in the unit bandwidths. The relative intensity noise is described the parameters of the quantum noise of the laser.

$$r_s = \frac{\langle \delta P_s^2 \rangle}{\langle P_s \rangle^2} = r_p \ln(G_R)$$

Mean square value of optical power jitter

Raman gain of the signal light

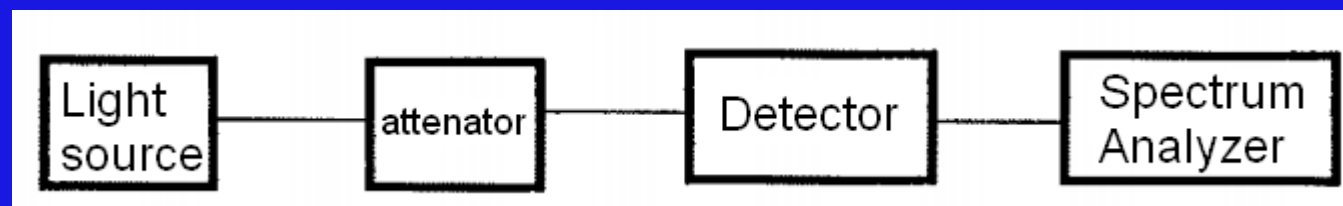
Mean value of optical power

$$r_s = r_p \cdot \ln(G_R)^2 \left(\frac{(V_s / L_{\text{eff}})^2}{(\alpha_p V_s)^2 + (2\pi D \cdot \Delta\lambda \cdot V_s f)^2} \right) \cdot \left(1 - 2e^{-\alpha_p L} \cos(2\pi D \cdot \Delta\lambda \cdot L f) + e^{-2\alpha_p L} \right)$$

Forward pump

$$r_s = r_p \cdot (\ln(G_R))^2 \frac{(V_s / L_{\text{eff}})^2}{(\alpha_p V_s)^2 + (4\pi f)^2} \cdot \left(1 - 2e^{-\alpha_p L} \cos(4\pi f T) + e^{-2\alpha_p L} \right)$$

Backward pump



Measure the RIN

Noise Figure

Definition

$$NF = \frac{SNR_{in}}{SNR_{out}} = \frac{N_{out}}{G \cdot N_{in}}$$

$$S_{shot} = 2h\nu GP_{in}$$

shot noise density

$$NF = 10 \log \left(\frac{S_{shot}}{2h\nu G^2 P_{in}} + \frac{S_{ex}}{2h\nu G^2 P_{in}} \right)$$

Excessive noise density

signal-spontaneous beat noise

spontaneous-spontaneous beat noise

Residual pump light caused noise

Net Noise Figure

ASE noise within B_0 , depending on pumping scheme

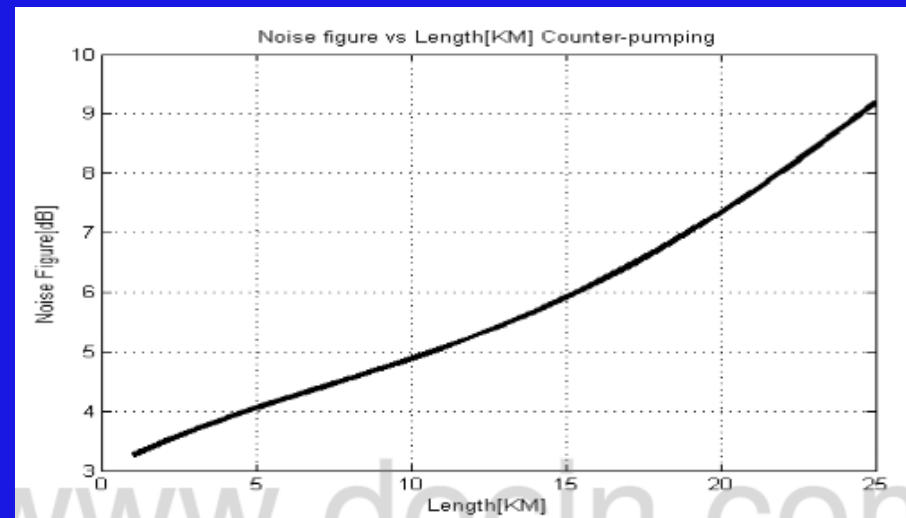
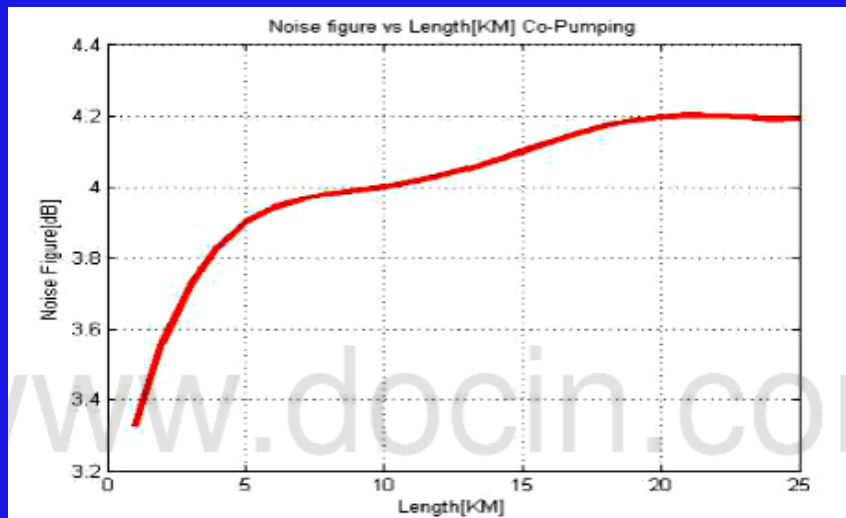
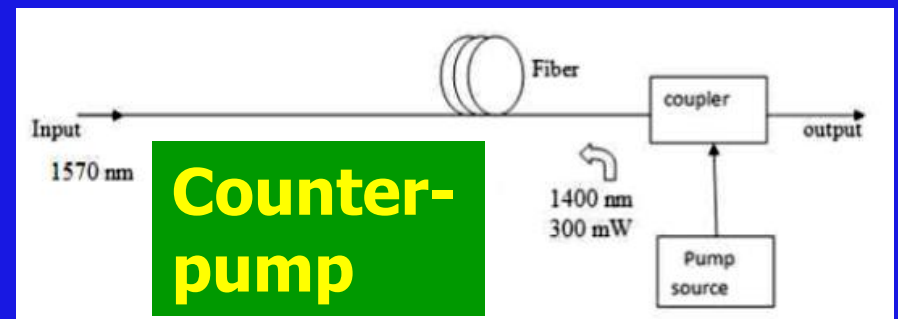
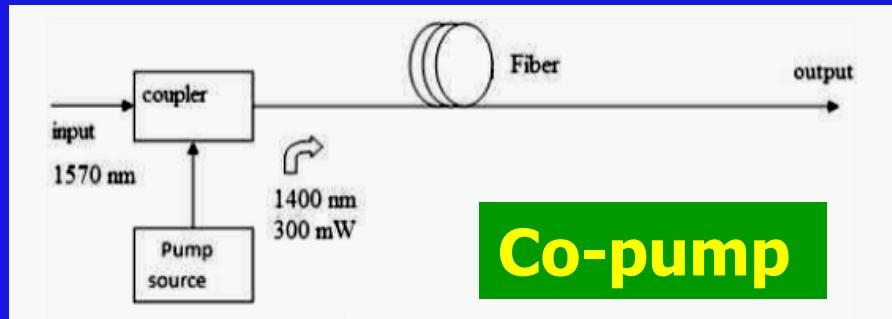
$$B_0 = \frac{c}{\lambda^2} \Delta\lambda = 6.2 \times 10^{10} \text{ Hz}$$

$$\Delta\lambda = 0.5 \text{ nm}$$

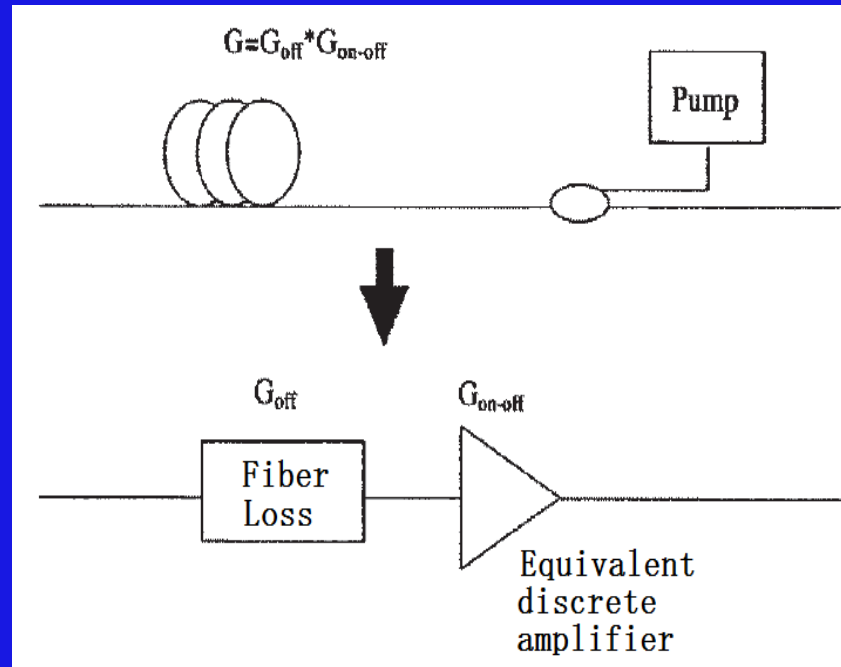
$$NF(\text{dB}) = 10 \log_{10} \left(\frac{P_{ASE}}{h\nu B_0 G} + \frac{1}{G} \right)$$

Gain

Noise Figure v.s. Pumping scheme

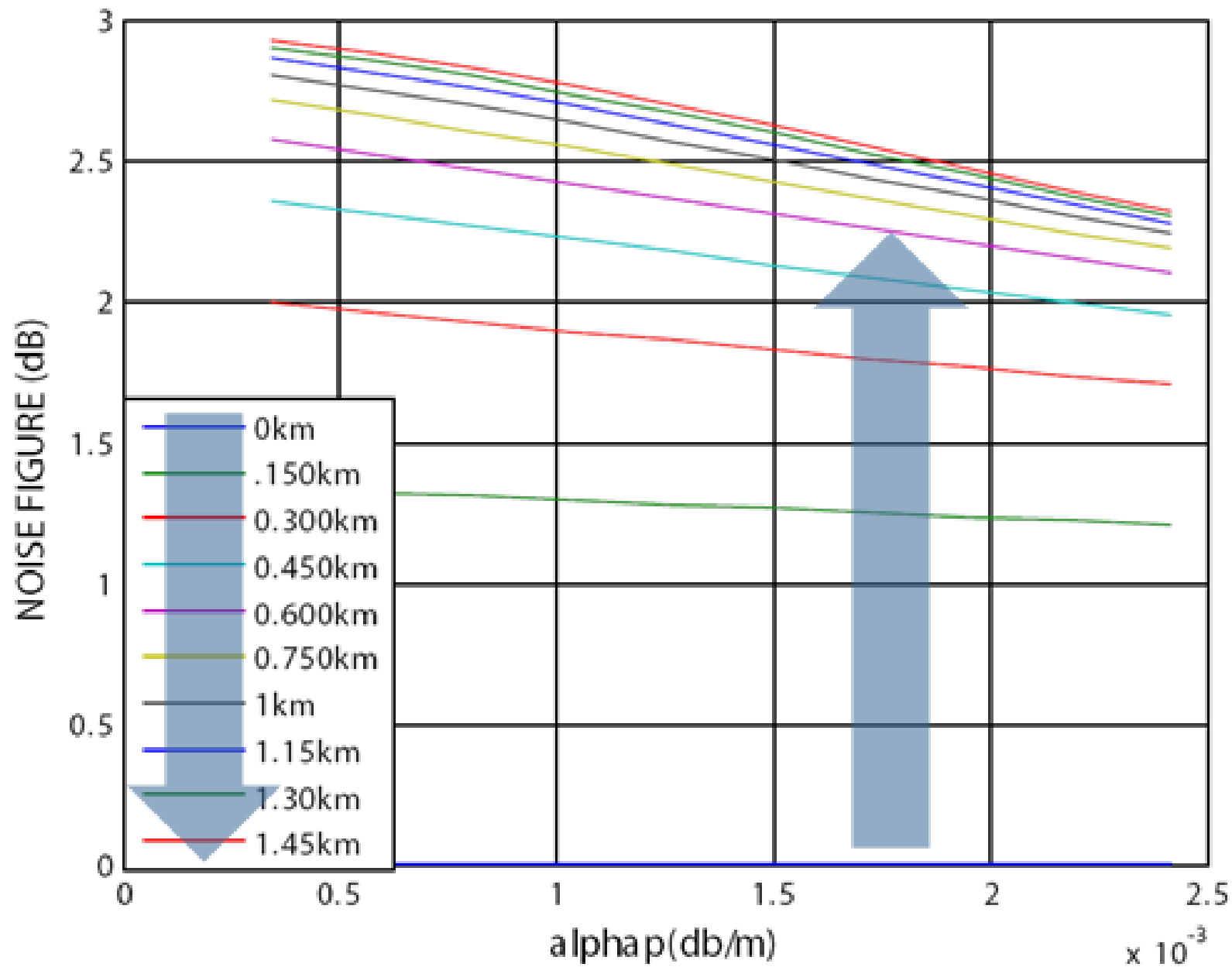


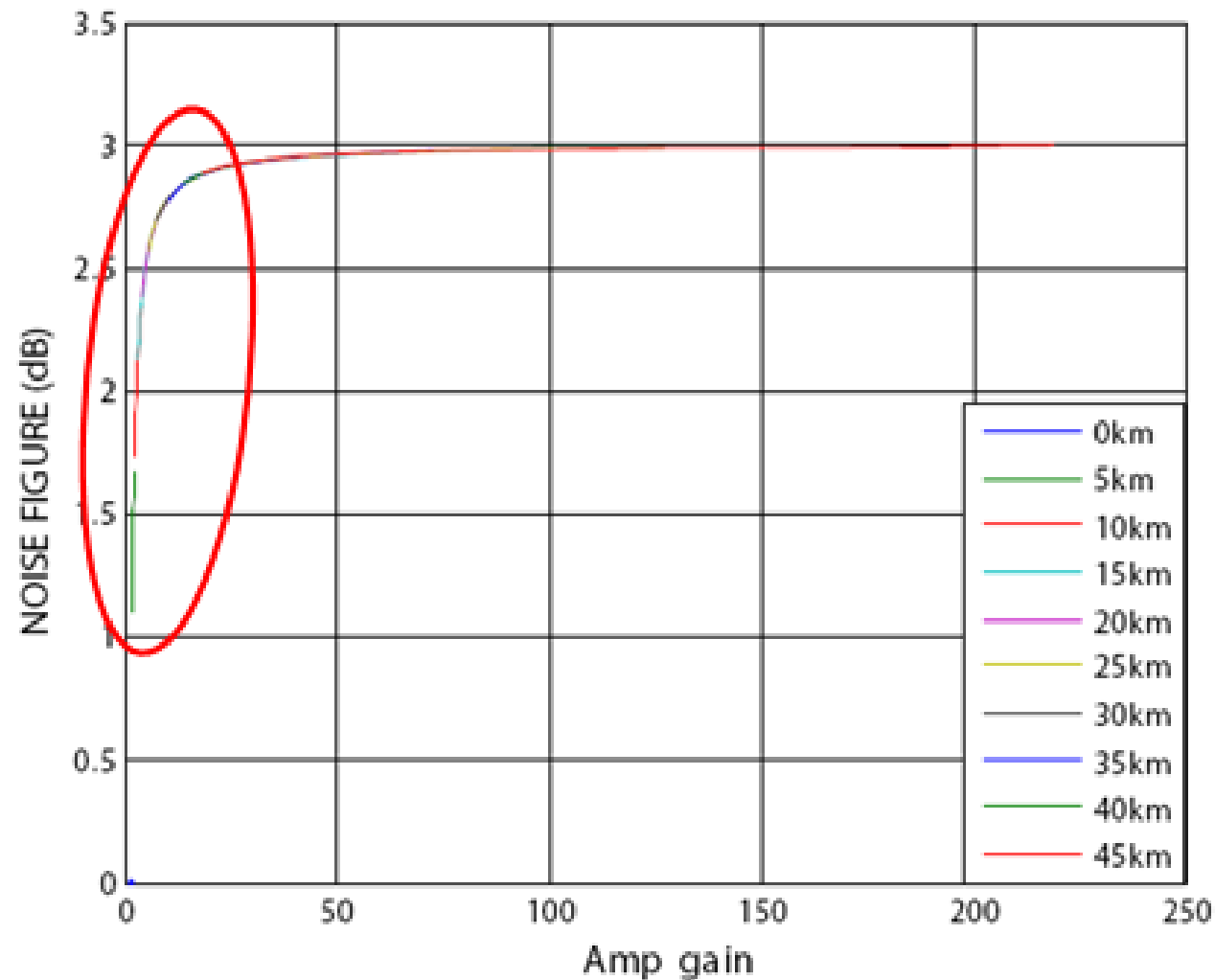
Equivalent Noise Figure



$$NF_{eq} = NF_{net} \cdot G_{off} = \frac{1}{G_{on-off}} \left(\frac{P_{ASE}}{h\nu \cdot B_0} + 1 \right)$$

$$NF_{eq} = 10 \log_{10} \left(\frac{P_{ASE}}{G_{on-off} h\nu \cdot B_0} + \frac{1}{G_{on-off}} \right)$$





Noise Fig. v/s Amplifier gain for various values of Amplifier length at $\text{GainR} = 1.02 \times 10^{-13}$ to 1.92×10^{-13} m/w.

In-class Exercise

Design a multi-pump Raman amplifier
Optimize/flatten gain by adjusting pumping powers

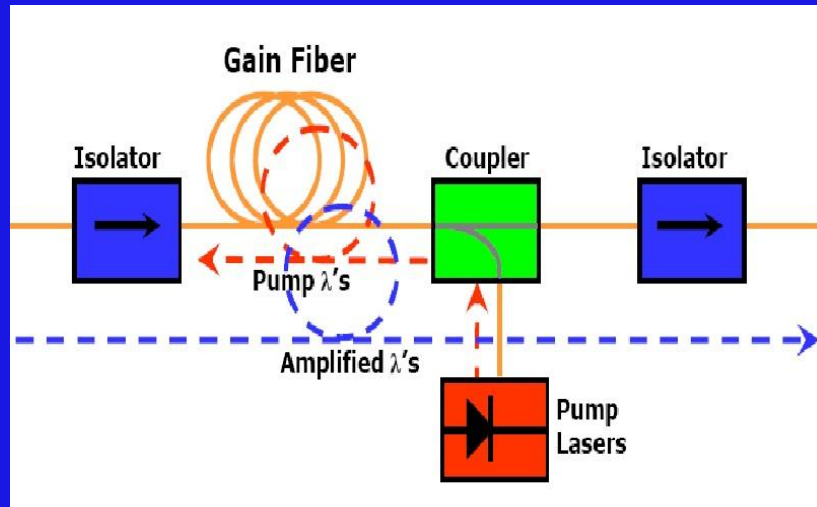


Table 1: Initial pump wavelength and power allocation

Wavelength (nm)	Power (mW)
1405.0	110.0
1415.0	140.0
1435.0	170.0
1460.0	140.0

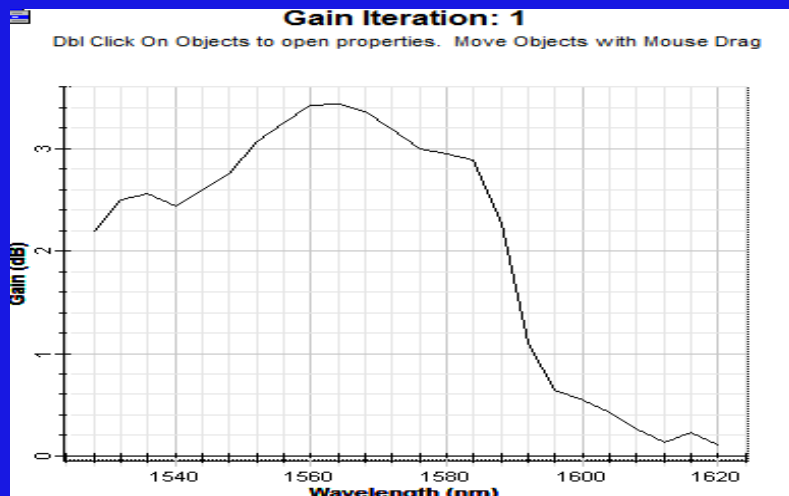
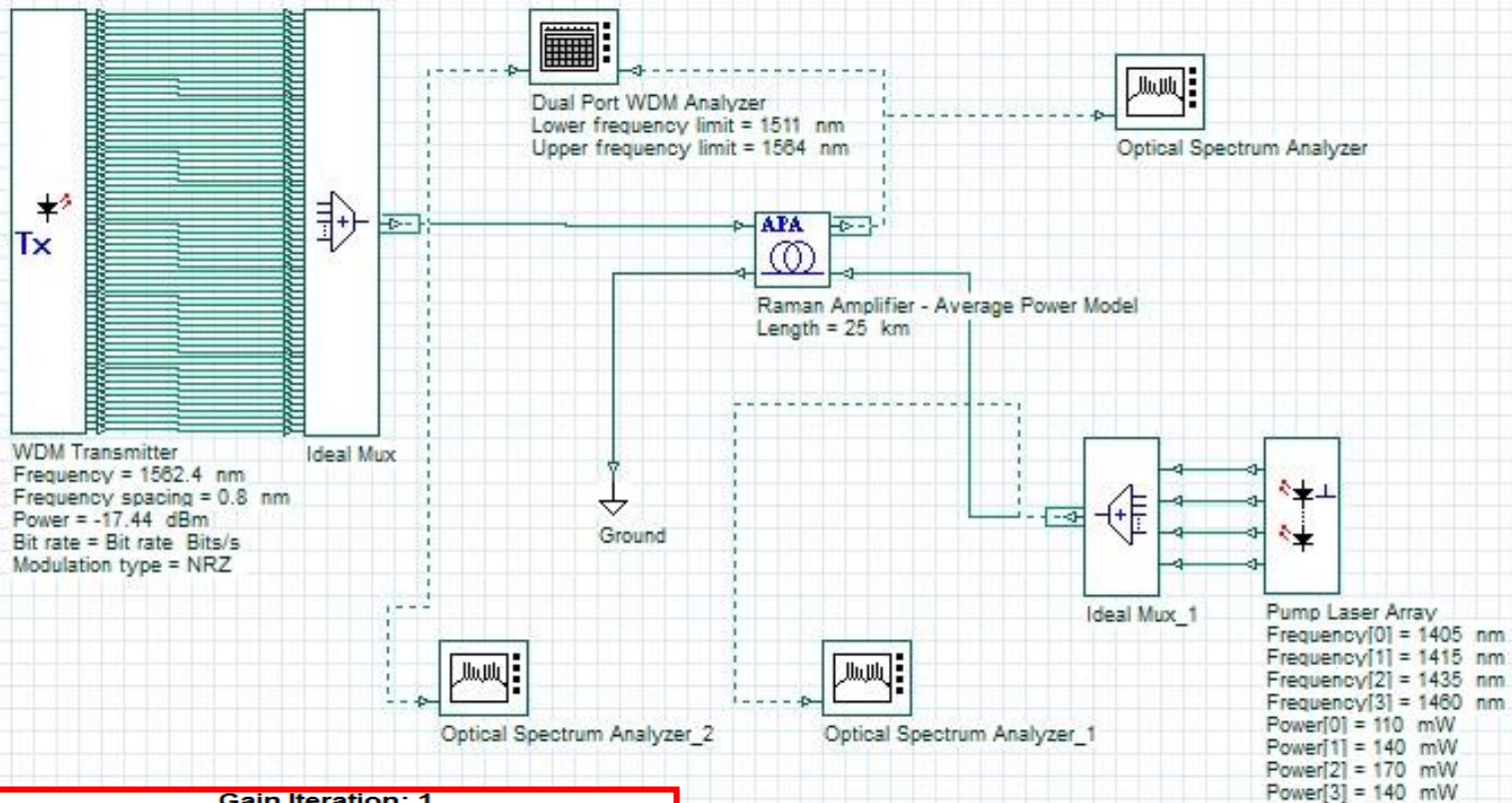


Table 2: Optimum pump wavelength and power allocation

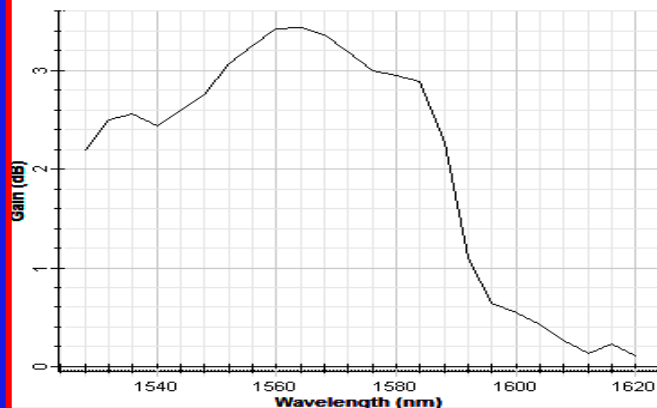
Wavelength (nm)	Power (mW)
1408.6	169.8
1419.9	204.1
1433.1	228.0
1473.4	198.0

In-class Exercise

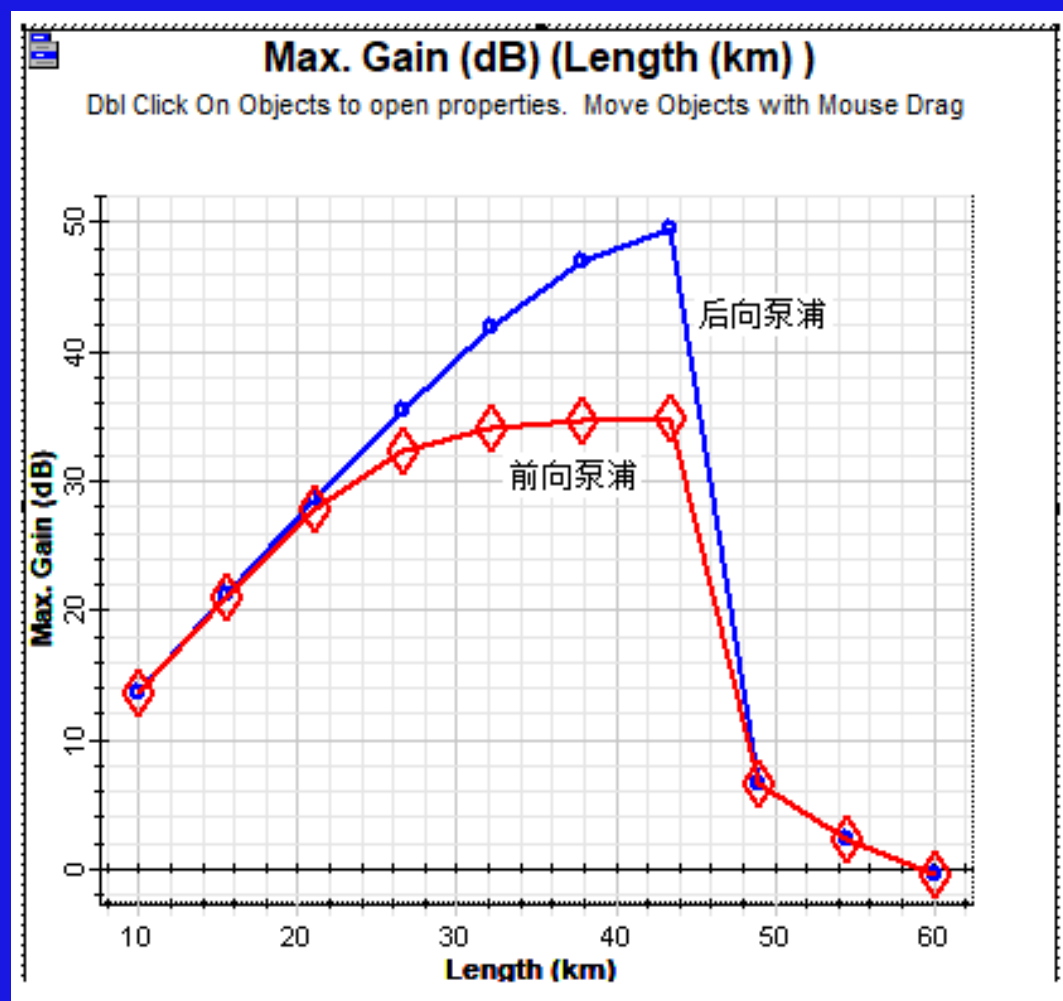


Gain Iteration: 1

Click On Objects to open properties. Move Objects with Mouse Drag



拉曼增益与拉曼光纤长度的关系



放大器长度扫描

